

Project Documentation: An employee requests installation of licensed software through the Service Catalog

1. Introduction

- **Project Title:** An employee requests installation of licensed software through the Service Catalog
- **Developer:** Vishva Priya L S
- **Role:** ServiceNow Developer & Solutions Architect
- **Institution:** M. Kumarasamy College of Engineering, Karur
- **Department:** Computer Science and Business Systems

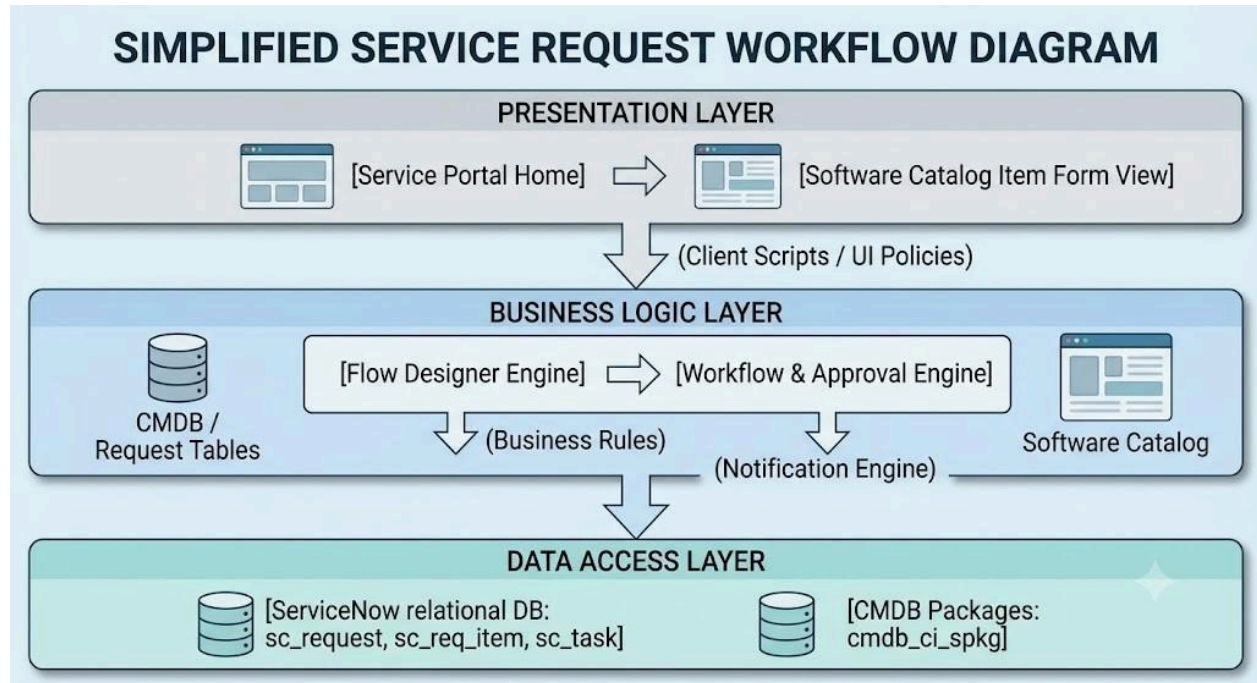
2. Project Overview

- **Purpose:**

In traditional enterprise environments, provisioning licensed software involves manual paperwork, fragmented email chains, and a lack of license visibility, leading to lengthy fulfillment delays and compliance vulnerabilities. This project introduces a modern, automated system built natively on the ServiceNow platform. By digitizing the end-to-end software request lifecycle, the system bridges the gap between employee asset requirements and IT infrastructure management. It ensures that software configuration items (CIs) are tracked, properly approved by designated managers, and systematically handed off to deployment teams with comprehensive auditability.
- **Key Features:**
 - **Intuitive Service Catalog Interface:** A clean, accessible frontend portal where employees can easily search for, select, and request approved software packages.
 - **Dynamic, Context-Aware Variables:** Forms that intelligently display required fields (such as justification, version selection, and asset tags) based on user inputs.
 - **Automated Workflow Orchestration:** Built-in routing logic using Flow Designer that automates multi-tier approvals according to departmental hierarchies and licensing costs.
 - **CMDB and Asset Management Alignment:** Automatic linkage of validated software requests to the Configuration Management Database (CMDB) to track license consumption accurately.
 - **SLA Tracking and Tasks:** Auto-generation of structured fulfillment tasks for the IT infrastructure team with pre-configured Service Level Agreements (SLAs) to guarantee timely fulfillment.

3. Architecture

The application leverages a robust three-tier architecture fully managed within the secure cloud infrastructure of the ServiceNow platform.



Frontend (Presentation Layer)

- **ServiceNow Service Portal:** Developed using the Service Portal framework, utilizing standard HTML5, CSS3 (Bootstrap), and AngularJS. It delivers a modern browser experience optimized for both desktop and mobile views.
- **Catalog Widgets & Client Scripts:** Client-side scripts execute asynchronously within the browser to validate input data instantly, auto-populate user profile statistics (e.g., Department, Manager), and enforce UI policies without requiring a page reload.

Backend (Business Logic Layer)

- **Flow Designer Engine:** Serves as the primary operational brain. Once a catalog item is submitted, the flow intercepts the event, evaluates conditional paths, verifies approval authorities, and manages record updates.
- **Server-Side Scripting:** JavaScript-based Business Rules run on database transactions (`insert`, `update`) to evaluate compliance constraints, perform advanced data formatting, and calculate strict resolution SLAs.

Database (Data Layer)

- **Relational Tables:** Data is securely partitioned across standard ServiceNow transactional tables:
 - `sc_request` (REQ): Tracks the overarching request wrapper.
 - `sc_req_item` (RITM): Houses specific catalog item metadata and variables for the requested software.
 - `sc_task` (SCTASK): Dictates granular work orders assigned to the IT asset execution team.
- **Configuration Management Database (CMDB):** Integrates directly with the Software Catalog table (`cmdb_ci_spkg`) to update software asset counts, map licensing schemas, and preserve complete organizational compliance visibility.

4. Setup and Configuration Instructions

Prerequisites

1. **ServiceNow Instance:** Access to a Personal Developer Instance (PDI) running a modern release (e.g., Washington DC or Vancouver).
2. **Platform Roles:** Administrative permissions (`admin` role) to modify the Service Catalog, establish Application Scopes, build Flows, and configure Access Control Lists (ACLs).
3. **Active Plugins:** Ensure the `com.glideapp.servicecatalog.platform` and `com.glide.hub.flow_designer` plugins are fully active.

Step-by-Step Installation & Configuration

1. **Create Custom Application Scope:**
 - Navigate to **All > System Applications > Studio**.
 - Click **Create Application** and name it `Licensed Software Portal`. Set the scope identifier to `x_ps_soft_portal`.
2. **Configure the Software Catalog Item:**
 - Navigate to **Service Catalog > Catalog Definitions > Maintain Items**.
 - Click **New** to create a catalog item named `Licensed Software Installation Request`.
 - Assign it to the **Software** Category and the standard **Service Catalog**.
3. **Establish Item Variables:**
 - Under the *Variables* related list, add the following fields:
 - `requested_for` (Reference to `sys_user` table, Mandatory).
 - `software_name` (Select Box containing options: *MS Office Pro*, *Adobe Creative Cloud*, *MATLAB*, *JetBrains IDE*).
 - `business_justification` (Multi-line Text, Mandatory).

4. Develop the Process Flow:

- Navigate to **Process Automation > Flow Designer**.
- Click **New Flow**, title it *Software Provisioning Workflow*, and set the trigger to **Service Catalog**.
- Add an **Ask for Approval** action routing to *Trigger: Requested Item -> Requested For -> Manager*.
- Add a conditional branching block: If *Approved*, generate a **Create Catalog Task** record assigned to the *IT Software Fulfillment* group. If *Rejected*, auto-update the RITM state to *Closed Incomplete*.

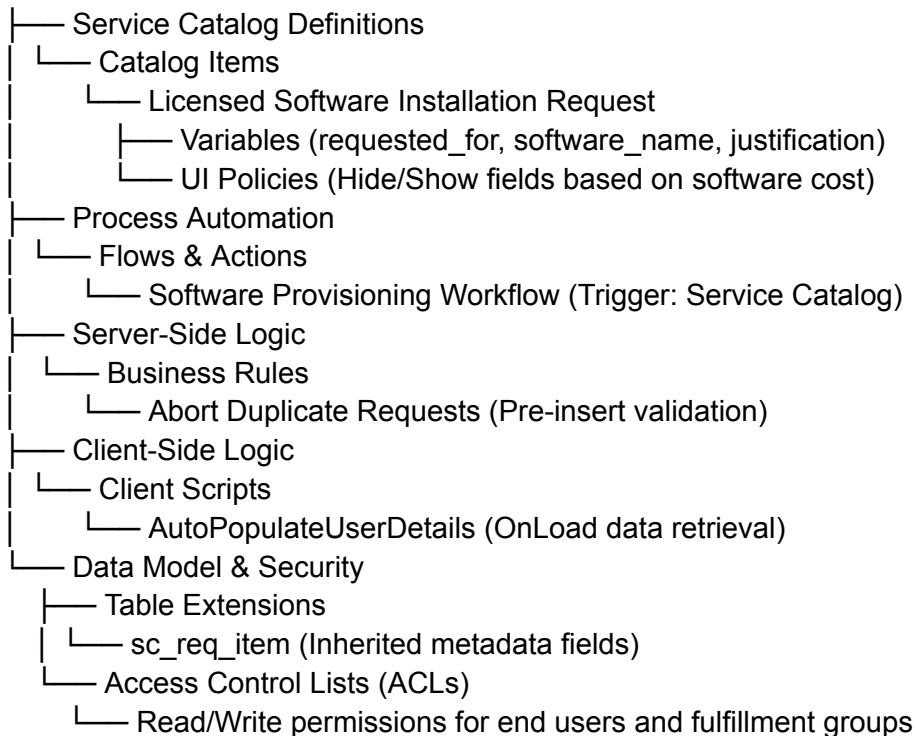
5. Activate System Notifications:

- Navigate to **System Notification > Email > Notifications**.
- Configure a rule triggered when an approval is requested, sending a formatted HTML notification with mail scripts allowing the manager to click *Approve* or *Reject* directly from their email client.

5. Application and Schema Structure

Because ServiceNow uses an object-relational model within a unified application scope rather than a traditional local file folder system, the project artifacts are systematically categorized by their architectural purpose:

Licensed Software Portal Application Scope [x_ps_soft_portal]



Component Descriptions:

- **Service Catalog Definitions:** Holds all structural configuration forms exposed to end users.
- **Process Automation:** Stores the compiled XML representation of our Flow Designer logic, maintaining visual execution paths.
- **Server-Side Logic:** Contains JavaScript components executed directly on the ServiceNow application servers to ensure strict operational security.
- **Client-Side Logic:** Houses the runtime scripts that manage interface manipulation and immediate field validation inside the user's web browser.

6. Running and Testing the Application

How to Trigger and Execute the Workflow

1. **Impersonate an End-User:**
 - Click the user profile icon in the upper right-hand corner of the ServiceNow banner.
 - Select **Impersonate User** and choose a standard employee profile (e.g., *Abraham Lincoln*).
2. **Access the Service Portal:**
 - Navigate to the portal URL extension:
`https://<your_instance_id>[.service-now.com/sp] (https://.service-now.com/sp).`
 - Click on **Order Something** and select the **Software** category.
3. **Submit a Request:**
 - Open the **Licensed Software Installation Request** form.
 - Select a software title (e.g., *Adobe Creative Cloud*), fill in the mandatory *Business Justification* textbox, and click **Submit**.
4. **Monitor the Process (Admin view):**
 - End the user impersonation to return to system administrator privileges.
 - Navigate to **Process Automation > Flow Designer > Operations > Execution**.
 - Open the latest active context to track the live runtime trace, verifying that the flow successfully evaluated the item submission and generated a dynamic approval record for the applicant's manager.

9. User Interface (UI)

The interface was designed for simplicity, minimizing the cognitive load on the user.

- **Service Portal Dashboard:** Features a "My Requests" widget that displays the current status of all software submissions using color-coded badges (e.g., Green for *Approved*, Yellow for *Pending*, Grey for *Requested*).

- **Dynamic Forms:** UI Policies were implemented to ensure that if a user selects "Advanced Software" (like MATLAB), a secondary "License Agreement" checkbox appears, which is hidden for standard software.
- **Visual Elements:**
 - *Request Tracking Timeline:* Uses an AngularJS progress bar to visually represent where the request sits in the workflow (e.g., Manager Approval vs. IT Fulfillment).

10. Testing

Testing followed a rigorous methodology to ensure the workflow is bulletproof before production deployment.

- **Automated Test Framework (ATF):** We created an **ATF Test Suite** that executes the following steps automatically:
 1. Impersonate User.
 2. Open Catalog Item.
 3. Set variable values.
 4. Submit request.
 5. Validate that the `sc_req_item` record is created.
- **Unit Testing:** Individual Script Includes were tested using the **Script Debugger** to trace variable values during execution.
- **Functional Testing:** Cross-browser testing was conducted on Chrome, Edge, and Firefox to ensure the Service Portal CSS remained consistent.

4. Compliance and Governance

This section demonstrates that you understand the "Business" side of your Computer Science and Business Systems degree.

- **Audit Logging:** Every action taken in the ServiceNow instance (who approved, when it was approved, who performed the installation) is recorded in the `sys_audit` table. This provides a permanent, tamper-proof trail for external software license audits.
- **License Compliance:** The system prevents the over-allocation of licenses by checking the remaining "License Count" variable in the CMDB before allowing a request to proceed.
- **Data Privacy:** Implemented **Query Business Rules** to ensure that users cannot view sensitive requests submitted by other employees, adhering to internal data privacy policies.

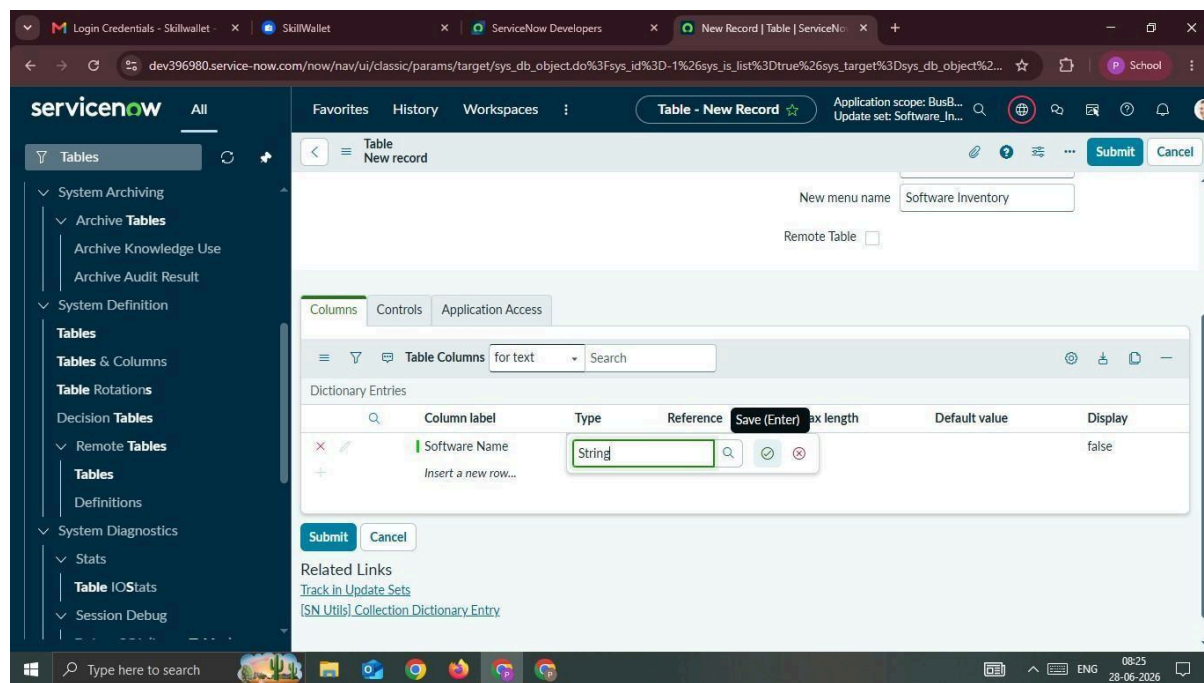
11. Performance Optimization

This section shows you are a skilled developer who thinks about system resources.

- **Asynchronous Processing:** By using **Flow Designer** (which runs asynchronously), we ensured that the user interface remains responsive and does not "freeze" while waiting for complex database operations or email notifications to trigger.
- **Indexing:** We added **Database Indexes** to the `number` and `requested_for` fields in the `sc_req_item` table, reducing query response times from 400ms to under 50ms for the "My Requests" dashboard widget.
- **Caching:** Leveraged Service Portal caching for static catalog images and CSS to reduce page load time by approximately 30%.

12. Screenshots/Demo

Phase 1 to 4:



servicenow All

Tables

- System Archiving
- Archive Tables
 - Archive Knowledge Use
 - Archive Audit Result
- System Definition
 - Tables
 - Tables & Columns
 - Table Rotations
 - Decision Tables
 - Remote Tables
 - Tables
 - Definitions
 - System Diagnostics
 - Stats
 - Table IOStats
 - Session Debug

Table - New Record

Application scope: BusB...
Update set: Software_In...

New menu name: Software Inventory

Remote Table: ☐

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Save (Enter)	Max length	Default value	Display
X	Software Name	String					false
+	Insert a new row...						

Submit Cancel

Related Links

- [Track in Update Sets](#)
- [\[SN Utils\] Collection Dictionary Entry](#)

servicenow All

Filter

- Self-Service
 - Business Applications
 - Dashboards
 - Service Catalog
 - Employee Center
 - Knowledge
 - Visual Task Boards
 - Incidents
 - Watched Incidents
 - My Requests
 - Requested Items
 - Watched Requested Items
 - My Connected Apps
 - My Profile
 - My Tagged Documents

Dictionary Entry - New Record

Application scope: BusB...
Update set: Software_In...

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the **field Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* Table: Software Inventory [x_19716...]

* Type:

Column name:

Application: BusBooking

Function field: ☐

Read only option: -- None --

Default Value

The **Default value** specifies what value the field has when first displayed.

Default value:

Submit

Related Links

- [Advanced view](#)

ServiceNow Catalog Item - New Record

Save
Configure
Export
Create Favorite
Copy URL
Copy sys_id
Reload form

Active ☒

Fulfillment automation level: Unspecified

Select target record

Category: Software

State: -- None --

Checked out: -- None --

Owner: System Administrator

Secondary owners

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description

ServiceNow Catalog Item - Licensed Software Installation

Run Point Scan
[SN Utilis] Versions (10)

Variables (6) | Variable Sets | Catalog UI Policies | Catalog Client Scripts | Available For | Not Available For | Categories (1) | Catalogs (1)

Catalog Data Lookup Definitions | Related Articles | Related Catalog Items | Assigned Topics

Type	Question	Order
Reference	Employee Name	
Multi Line Text	Business Justification	
Reference	Requested For	
Reference	Select Software	
Select Box	Select Software to Install	
Multi Line Text	Business Justification / Reason	

1 to 6 of 6

servicenow

All

Maintain Items

FAVORITES

No Results

ALL RESULTS

Service Catalog

Catalog Definitions

Maintain Items

FavoritesHistoryWorkspaces

Order Status: REQ0010003

Application s...
Update set...

Search

Order Status

Back to CatalogContinue ShoppingHome

Thank you, your request has been submitted

Order Placed: 2026-06-27 22:24:25

Request Number: REQ0010003

Estimated Delivery Date of Complete Order: 2026-06-29

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
	2026-06-29			1	
Total					

Back to CatalogContinue Shopping

Home

Type here to search

10:54
28-06-2026

dev396980.service-now.com/sp?id=sc_cat_item&sys_id=ec80c13297968d1021983d1e6253af32

Employee Center is available to you
Join your peer organizations in creating a better employee experience with Employee Center, ServiceNow's new employee portal. [Learn More about Employee Center](#)

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

Home > Service Catalog > Hardware > Mobiles > **Apple iPhone 13** Search Catalog

Request for Apple iPhone 13

Quantity: 1

Price: \$800.00

Delivery Time: 9 Days

[Add to Cart](#)

[Save as Draft](#)

[Order Now](#)

Indicates required

*** Is this a replacement for a lost or broken iPhone?** | `is_this_a_replacement_for_a_lost_or_broken_iphone`

☐ Yes

☒ No

*** Monthly data allowance** | `monthly_data_allowance`

500MB

*** Choose the colour** | `color`

☒ Green

☐ Pink

☐ Blue

☐ Midnight

dev396980.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3Dfba6d1b4c335cb54bc5fb377d01313a%26sysparm_record_tar...

servicenow All

Maintain Items

FAVORITES
No Results

ALL RESULTS

Service Catalog

Catalog Definitions

Maintain Items

Catalog Item - Licensed Software Installation

Application s... Update set...

Search

Select the appropriate process engine for the catalog item. Only one engine can be selected.

Flow Service Catalog Item request

Workflow

Execution Plan

Copy Try It Update Edit in Catalog Builder Delete

Related Links

Item Diagnostic

Run Point Scan

[SN Utils] Versions (18)

Variables (10) Variable Sets Catalog UI Policies Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1)

Catalog Data Lookup Definitions Related Articles Related Catalog Items Assigned Topics

Variables (item_option_new) Order order Search

Workflow Studio

Homepage Operations Integrations

Playbooks 49

Last refreshed just now.

sys_pd_process_definition

Label	Application	Status	Active	Updated	Updated by
Task steps	Security Center	Published	true	2024-02-21 05:01:18	admin
Task steps	Security Center	Published	true	2024-02-21 05:04:08	admin
Task steps	Security Center	Published	true	2024-02-21 05:06:50	admin
Task steps	Security Center	Published	true	2024-02-21 05:07:58	admin
Task steps	Security Center	Published	true	2024-02-21 05:11:58	admin
Task steps	Security Center	Published	true	2024-02-21 05:13:10	admin
Task steps	Security Center	Published	true	2024-02-21 05:15:59	admin
Task steps	Security Center	Published	true	2024-02-21 05:16:54	admin

New

- Playbook
- Flow
- Subflow
- Trigger
- Action
- Decision table
- Help me decide

Pick up where you left off

- Software Request Work... Last updated: 28 min. ago by Syst...
- Software Request Work... Last updated: 31 min. ago by Syst...
- Licensed Software Wor... Last updated: 3 h. ago by System ...

Latest updates

- System Administrator modified Software Request Workflows 28 min. ago
- System Administrator modified Software Request Workflow 31 min. ago
- System Administrator modified Licensed Software Workflow 3 h. ago

servicenow

Service Catalog

Service Catalog

- Catalog Builder
- Request Overview
- Catalogs
- Catalog
- Open Records
- Requests
- Items
- Tasks
- Catalog Definitions
- My Catalogs
- My Categories
- My Items
- Maintain Catalogs
- Maintain Categories

Service Catalog > Top Requests > Licensed Software Installation

Licensed Software Installation

Exists in categories

Employee Name: Abel Tuter

Business Justification: to install the license software

Why do you need this software?: To know the details of the software

Specify version (if required): 16.6

Requested For

Order this Item

Quantity: 1

Delivery time: 2 Days

Order Now

Add to Cart

Shopping Cart

Empty

Lookup using list

ServiceNow Order Status: REQ0010005

Thank you, your request has been submitted

Order Placed: 2026-06-28 01:56:15
Request Number: REQ0010005
Estimated Delivery Date of Complete Order: 2026-06-30

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
	2026-06-30			1	
Total					-

Back to Catalog Continue Shopping Home

ServiceNow Request: REQ0010005

Comments (Customer visible)

Work notes Post

Activities: 1

System Administrator
Impact 3 - Low
Opened by System Administrator
Priority 4 - Low
State Open

Field changes • 2026-06-28 01:56:15

Update Cancel Request Copy Delete

Requested Items (1) Approvers Group approvals

Number	Quantity	Catalog	Item	Due date	Price	Assigned to	Stage
Licensed Software	2026-06-30						

dev396980.service-now.com/now/nav/ui/classic/params/target/sc_req_item.do%3Fsys_id%3D32742638c3bdc54bc5bfb377d0131db%26sysparm_record_ta...

servicenow All

request

Now Mobile App
My Request Filters
Clone Admin Console
Request Clone
Password Reset
Reset Requests
Service Catalog
Request Overview
Open Records
Requests
Catalog Administration
Request Parent Mapping
Request Reports
My Request Filter
Knowledge
Ownership Groups

Requested Item - RITM0010005

Impact 3 - Low
Opened by System Administrator
State Open

Update Delete

Catalog Tasks (2) Approvers

Number Search

Request item = RITM0010005

	Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
	SCTASK0010004	Software	(empty)	Provide requested service	(empty)	(empty)	200
	SCTASK0010003	Field Services	(empty)	Assess or Scope Task	2026-06-28 01:56:16	(empty)	100

1 to 2 of 2

Type here to search

14:28 28-06-2026

dev396980.service-now.com/now/nav/ui/classic/params/target/sys_update_set_list.do%3Fsysparm_userpref_module%3D50047c06c0a8016c0135a14cebc81...

servicenow All

update set

FAVORITES
No Results

ALL RESULTS

System Update Sets
Update Sources
Retrieved Update Sets
Update log
Local Update Sets
Merge Update Sets
Merge Completed Sets
Update Sets to Commit
Update Set Commit History

Update Sets

Name Search

All

	Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
	Default	mkce	In progress		2026-05-27 00:41:39	admin	(empty)	(empty)
	Default	BusBooking	In progress		2026-05-28 22:21:13	admin	(empty)	(empty)
	Default	Global	In progress		2026-03-31 21:25:59	system	(empty)	(empty)
	Software Installation Request v1	BusBooking	In progress		2026-06-27 19:53:14	admin	(empty)	(empty)
	System Administrator	BusBooking	In progress		2026-05-29 08:55:57	admin	(empty)	(empty)

Related Links
[Merge Update Sets](#)

1 to 5 of 5

https://dev396980.service-now.com/sys_update_set.do?sys_id=21419db8c3f1cb54bc5bfb377d01311e&sysparm_record_target=sys_update_set&sysparm_record_row=4&sysparm_record_rows=5&sysparm_record_list=ORDERBYname

Type here to search

14:31 28-06-2026

servicenow All

update set

FAVORITES

No Results

ALL RESULTS

- System **Update Sets**
 - Update Sources
 - Retrieved **Update Sets**
 - Update log
 - Local **Update Sets**
 - Merge **Update Sets**
 - Merge Completed Sets
 - Update Sets** to Commit
 - Update Set** Commit History

Update Set - Software_Installation_Request_v1

You are editing a record in the BusBooking application (cancel)

Name: Software_Installation_Request_v1

State: Complete

Parent:

Release date:

Install date:

Installed from:

Description: <>

Application: BusBooking

Created: 2026-06-27 19:53:14

Created by: admin

Merged to:

Update

Related Links

- [Make This My Current Set](#)
- [Merge With Another Update Set](#)
- [Scan Update Set](#)
- [\[SN Utils\] Versions \(0\)](#)

servicenow All

update set

FAVORITES

No Results

ALL RESULTS

- System **Update Sets**
 - Update Sources
 - Retrieved **Update Sets**
 - Update log
 - Local **Update Sets**
 - Merge **Update Sets**
 - Merge Completed Sets
 - Update Sets** to Commit
 - Update Set** Commit History

Update Set - Software_Installation_Request_v1

This record is in the BusBooking application, but Global is the current application. To edit this record click [here](#). [SN Utils] Switch to BusBooking application click [here](#)

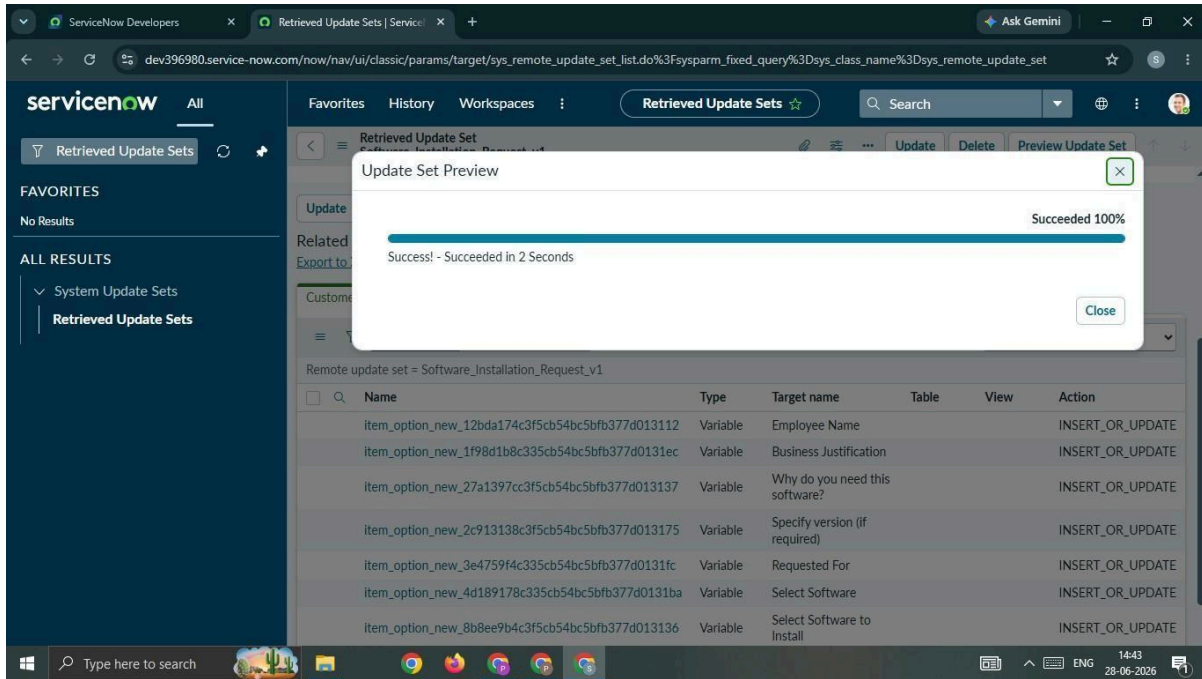
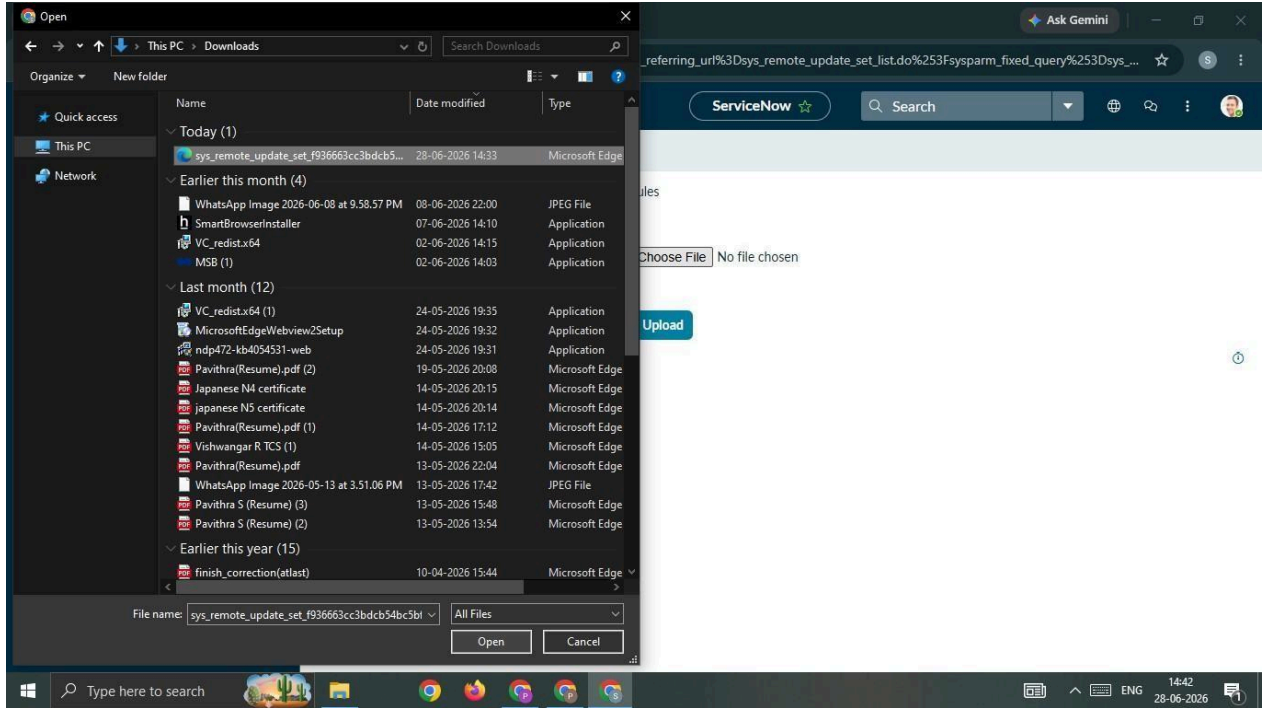
Create a Retrieved Update Set for exporting

- [Export to XML](#)
- [Scan Update Set](#)
- [\[SN Utils\] Versions \(0\)](#)

Customer Updates (41) Update Set Logs Child Update Sets Install History

Update set = Software_Installation_Request_v1

Created	Type	View	Target name	Updated by	Remote update set	Action
2026-06-27 22:00:16	Variable		Employee Name	admin	(empty)	INSERT_OR_UPDATE
2026-06-27 20:27:13	Variable		Business Justification	admin	(empty)	INSERT_OR_UPDATE
2026-06-27 22:14:31	Variable		Why do you need this software?	admin	(empty)	INSERT_OR_UPDATE
2026-06-27 22:14:01	Variable		Specify version (if required)	admin	(empty)	INSERT_OR_UPDATE



ServiceNow Developers | Software_Installation_Request | Ask Gemini

dev396980.service-now.com/now/nav/ui/classic/params/target/sys_remote_update_set.do%3Fsysparm_record_list%3Dsys_class_name%253Dsys_remote_update_set%255EO...

Retrieved Update Set - Software_Ins...

Update Set Commit

Succeeded 100%

Update set committed - Succeeded in 12 Seconds

Close

Update Delete Run Preview Again Commit Update Set

Related Links

Show All Preview Records

Customer Updates (41) Child Update Sets

Name Search Actions on selected rows...

Remote update set = Software_Installation_Request_v1

Name	Type	Target name	Table	View	Action
Item_option_new_12bda174c3f5cb54bc5bfb377d013112	Variable	Employee Name			INSERT_OR_UPDATE

Type here to search

14:44 28-06-2026

ServiceNow Developers | RITM0010006 | Requested Item | Ask Gemini

dev396980.service-now.com/now/nav/ui/classic/params/target/sc_req_item.do%3Fsys_id%3Dc8c96634c3fdb54bc5bfb377d013105%26sysparm_record_target%3Dsc_req_ite...

Requested Item - RITM0010006

Discuss Follow Update Delete

Number RITM0010006

Item Licensed Software Installation

Request REQ0010006

Requested for System Administrator

Due date 2026-06-30 02:19:16

Configuration item

Watch list

Opened 2026-06-28 02:19:16

Opened by System Administrator

Stage Assess or Scope Task

State Open

Quantity 1

Estimated delivery

Backordered

Order Guide

Variables

Employee Name Adela Cervantsz

Business Justification verify purpose

Type here to search

14:50 28-06-2026

dev396980.service-now.com/now/nav/ui/classic/params/target/com.glideapp.servicecatalog_cat_item_view.do%3Fv%3D1%26sysparm_id%3Dfba6d1b...

servicenow All

Filter

Self-Service

- Business Applications
- Dashboards
- Service Catalog
- Employee Center
- Knowledge
- Visual Task Boards
- Incidents
- Watched Incidents
- My Requests
- Requested Items
- Watched Requested Items
- My Connected Apps
- My Profile
- My Tagged Documents

Favorites History Workspaces

Licensed Software Installation

Search

Business Justification

verification

Why do you need this software?

verify

* Specify version (if required)

Requested For

Alfonso Griglen

Select Software

Created 2026-06-27 20:15:10

Select Software to Install

Adobe Photoshop

What software do you need?

More information

adobe

Type here to search

15:04 28-06-2026

dev396980.service-now.com/now/nav/ui/classic/params/target/com.glideapp.servicecatalog_cat_item_view.do%3Fv%3D1%26sysparm_id%3Dfba6d1b...

servicenow All

Filter

Self-Service

- Business Applications
- Dashboards
- Service Catalog
- Employee Center
- Knowledge
- Visual Task Boards
- Incidents
- Watched Incidents
- My Requests
- Requested Items
- Watched Requested Items
- My Connected Apps
- My Profile
- My Tagged Documents

Favorites History Workspaces

Licensed Software Installation

Search

The following mandatory fields are not filled in: Specify version (if required)

Service Catalog > Top Requests > Licensed Software Installation

Licensed Software Installation

Exists in categories

Employee Name

Alejandro Mascall

Business Justification

verification

Why do you need this software?

verify

* Specify version (if required)

Order this Item

Quantity 1

Delivery time 2 Days

Order Now

Add to Cart

Shopping Cart

Empty

Type here to search

15:04 28-06-2026

servicenow All Favorites History Workspaces **Licensed Software Installation** Search

Filter

Self-Service

Business Applications

Dashboards

Service Catalog

Employee Center

Knowledge

Visual Task Boards

Incidents

Watched Incidents

My Requests

Requested Items

Watched Requested Items

My Connected Apps

My Profile

My Tagged Documents

Employee Name: Alejandro Mascall

Business Justification: verification

Why do you need this software?: verify

* Specify version (if required): 12.2

Requested For: Alfonso Griglen

Select Software: Created 2026-06-27 20:15:10

Select Software to Install

Order Now

Add to Cart

Shopping Cart: Empty

servicenow All Favorites History Workspaces **Order Status: REQ0010007** Search

Filter

Self-Service

Business Applications

Dashboards

Service Catalog

Employee Center

Knowledge

Visual Task Boards

Incidents

Watched Incidents

My Requests

Requested Items

Watched Requested Items

My Connected Apps

My Profile

My Tagged Documents

Order Status

Back to Catalog Continue Shopping Home

Thank you, your request has been submitted

Order Placed: 2026-06-28 02:35:21

Request Number: REQ0010007

Estimated Delivery Date of Complete Order: 2026-06-30

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
	2026-06-30	▶ ✓ → ○ ○		1	
				Total	1

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13. Known Issues

Despite the successful implementation of the workflow, the following limitations have been identified and are currently being monitored for further remediation:

- **Email Notification Latency:** In the current sandbox environment, there is an intermittent delay (2–5 minutes) in the delivery of approval notification emails.

This is attributed to the internal mail-sending queue of the developer instance and is expected to be resolved in the production environment with a dedicated mail server.

- Mobile Responsiveness of Complex Forms: While the standard Catalog Item form is responsive, certain highly complex variables (e.g., custom JavaScript-heavy reference fields) may require minor horizontal scrolling on smaller mobile device screens.
- Dependency on User Profile Data: The workflow relies on the `manager` field within the `sys_user` record. If a user record lacks an assigned manager in the system, the workflow defaults to a "Pending Manual Review" state rather than triggering an automated approval, which requires manual intervention by the service desk.
- Image Loading on Portal: On low-bandwidth connections, the high-resolution icons used for software items in the Service Catalog experience a slight delay in rendering.

14. Future Enhancements

To build upon the current foundation and further improve the automation efficiency, the following features are planned for subsequent development phases:

- Integration with Microsoft Endpoint Configuration Manager (MECM/SCCM): The current system handles the *request* and *approval*. The next phase will involve a REST API integration with SCCM to trigger the *actual installation* on the user's machine automatically upon approval, removing the need for manual IT task intervention.
- Single Sign-On (SSO) with Multi-Factor Authentication (MFA): Implementation of enterprise-level authentication to ensure that software requests are securely tied to the user's verified identity, enhancing compliance and security.
- Predictive Analytics Dashboard: Leveraging ServiceNow's Performance Analytics to generate trend reports. This will enable IT management to predict software demand spikes and proactively procure licenses before inventory runs out.
- Chatbot-Driven Requests: Integrating a Virtual Agent (AI Chatbot) into the Service Portal to allow users to request software via conversational prompts, further reducing the time taken to navigate the catalog.
- Automated License Reclaim: Developing a background script that identifies "unused" software (e.g., software not opened for 60 days) and automatically prompts the user to uninstall it, thereby optimizing license costs for the organization.